Simple Identities

Definition: $\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} f(x)\, p_X(x)\, \mathrm{d}x$, and for two variables, $\mathbb{E}[h(U,W)] = \iint h(u,w)\, p_{U,W}(u,w)\, \mathrm{d}u\, \mathrm{d}w$.

1 $\mathbb{E}[f(W) + g(U)] = \mathbb{E}[f(W)] + \mathbb{E}[g(U)]$

$$\begin{aligned}
\mathbb{E}[f(W) + g(U)] &= \iint \big(f(w) + g(u)\big)\, p_{U,W}(u,w)\, \mathrm{d}u\, \mathrm{d}w \\
&= \int f(w) \left[\int p_{U,W}(u,w)\, \mathrm{d}u\right] \mathrm{d}w + \int g(u) \left[\int p_{U,W}(u,w)\, \mathrm{d}w\right] \mathrm{d}u \\
&= \int f(w)\, p_W(w)\, \mathrm{d}w + \int g(u)\, p_U(u)\, \mathrm{d}u \\
&= \mathbb{E}[f(W)] + \mathbb{E}[g(U)]
\end{aligned}$$

2 $\mathbb{E}[X + a] = \mathbb{E}[X] + a$

$$\mathbb{E}[X + a] = \int (x + a)\, p_X(x)\, \mathrm{d}x = \int x\, p_X(x)\, \mathrm{d}x + a \int p_X(x)\, \mathrm{d}x = \mathbb{E}[X] + a$$

3 $\mathbb{E}[aX] = a\mathbb{E}[X]$

$$\mathbb{E}[aX] = \int ax\, p_X(x)\, \mathrm{d}x = a \int x\, p_X(x)\, \mathrm{d}x = a\mathbb{E}[X]$$

4 $\mathrm{var}(aX) = a^2\, \mathrm{var}[X]$

$$\mathrm{var}(aX) = \mathbb{E}\big[(aX - a\mathbb{E}[X])^2\big] = \int a^2 (x - \mathbb{E}[X])^2\, p_X(x)\, \mathrm{d}x = a^2\, \mathrm{var}[X]$$

5 $\mathrm{var}(X + Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2C_{XY}$

Let $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$, so $\mathbb{E}[X + Y] = \mu_X + \mu_Y$.

$$\begin{aligned}
\mathrm{var}(Z) &= \mathbb{E}\Big[\big((X - \mu_X) + (Y - \mu_Y)\big)^2\Big] \\
&= \mathbb{E}\big[(X - \mu_X)^2\big] + \mathbb{E}\big[(Y - \mu_Y)^2\big] + 2\mathbb{E}\big[(X - \mu_X)(Y - \mu_Y)\big] \\
&= \mathrm{var}(X) + \mathrm{var}(Y) + 2C_{XY}
\end{aligned}$$

Sum of Two Independent Gaussian Random Variables

Let $X \sim \mathcal{N}(0, \sigma_1^2)$ and $Y \sim \mathcal{N}(0, \sigma_2^2)$ be independent, and let $Z = X + Y$. Then

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma_1} \exp\!\left(-\frac{x^2}{2\sigma_1^2}\right), \qquad f_Y(y) = \frac{1}{\sqrt{2\pi}\,\sigma_2} \exp\!\left(-\frac{y^2}{2\sigma_2^2}\right).$$

Since X and Y are independent, the PDF of their sum is the convolution of their PDFs:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx.$$

We will use the Gaussian integral (a Gaussian PDF integrates to 1):

$$\int_{-\infty}^{\infty} \exp\!\left(-\frac{u^2}{2\tau^2}\right) du = \sqrt{2\pi}\,\tau, \qquad \tau > 0. \tag{1}$$

Substituting the PDFs,

$$f_Z(z) = \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} e^{-E}\, dx, \qquad E = \frac{x^2}{2\sigma_1^2} + \frac{(z-x)^2}{2\sigma_2^2}.$$

Combining over a common denominator,

$$E = \frac{\sigma_2^2 x^2 + \sigma_1^2 (z-x)^2}{2\sigma_1^2\sigma_2^2} = \frac{(\sigma_1^2 + \sigma_2^2)\, x^2 - 2\sigma_1^2 z x + \sigma_1^2 z^2}{2\sigma_1^2\sigma_2^2}.$$

Let $s^2 = \sigma_1^2 + \sigma_2^2$ and $a = \sigma_1^2/s^2$. Then

$$E = \frac{s^2}{2\sigma_1^2\sigma_2^2} \left[x^2 - 2azx + az^2\right].$$

Completing the square, $x^2 - 2azx = (x - az)^2 - a^2 z^2$, so

$$x^2 - 2azx + az^2 = (x - az)^2 + a(1-a)\, z^2, \qquad a(1-a) = \frac{\sigma_1^2}{s^2} \cdot \frac{\sigma_2^2}{s^2} = \frac{\sigma_1^2\sigma_2^2}{s^4}.$$

Therefore

$$E = \frac{(x - az)^2}{2\tau^2} + \frac{z^2}{2s^2}, \qquad \tau^2 = \frac{\sigma_1^2\sigma_2^2}{s^2}.$$

The z^2 term does not depend on x, so

$$f_Z(z) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\!\left(-\frac{z^2}{2s^2}\right) \int_{-\infty}^{\infty} \exp\!\left(-\frac{(x-az)^2}{2\tau^2}\right) dx.$$

By (1) (the shift by az does not change the integral), the integral equals $\sqrt{2\pi}\,\tau = \sqrt{2\pi}\,\sigma_1\sigma_2/s$. Hence

$$f_Z(z) = \frac{1}{2\pi\sigma_1\sigma_2} \cdot \frac{\sqrt{2\pi}\,\sigma_1\sigma_2}{s} \exp\!\left(-\frac{z^2}{2s^2}\right) = \boxed{\frac{1}{\sqrt{2\pi}\,s} \exp\!\left(-\frac{z^2}{2s^2}\right), \quad s^2 = \sigma_1^2 + \sigma_2^2.}$$

So $Z \sim \mathcal{N}(0, \sigma_1^2 + \sigma_2^2)$: the sum is Gaussian. ■

Method 2: Fourier domain

We use the Fourier transform $\mathcal{F}\{f\}(\omega) = \int_{-\infty}^{\infty} f(x)\, e^{-i\omega x}\, dx$ and the **convolution theorem**:

$$\mathcal{F}\{f_X * f_Y\} = \mathcal{F}\{f_X\} \cdot \mathcal{F}\{f_Y\}.$$

Fourier transform of a Gaussian. For $f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-x^2/2\sigma^2}$,

$$\mathcal{F}\{f\}(\omega) = \frac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{\infty} \exp\!\left(-\frac{x^2}{2\sigma^2} - i\omega x\right) dx.$$

Completing the square in the exponent,

$$-\frac{x^2}{2\sigma^2} - i\omega x = -\frac{(x + i\sigma^2\omega)^2}{2\sigma^2} - \frac{\sigma^2\omega^2}{2},$$

which can be checked by expanding $(x + i\sigma^2\omega)^2 = x^2 + 2i\sigma^2\omega x - \sigma^4\omega^2$. Thus

$$\mathcal{F}\{f\}(\omega) = e^{-\sigma^2\omega^2/2} \cdot \frac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{\infty} \exp\!\left(-\frac{(x + i\sigma^2\omega)^2}{2\sigma^2}\right) dx = e^{-\sigma^2\omega^2/2},$$

since the remaining integral equals $\sqrt{2\pi}\,\sigma$ by (1) (the imaginary shift does not change its value, by a contour integration argument). The Fourier transform of a Gaussian is a Gaussian:

$$\mathcal{F}\{\mathcal{N}(0, \sigma^2)\}(\omega) = e^{-\sigma^2\omega^2/2}.$$

Product of the transforms. By the convolution theorem,

$$\mathcal{F}\{f_Z\}(\omega) = e^{-\sigma_1^2\omega^2/2}\, e^{-\sigma_2^2\omega^2/2} = e^{-(\sigma_1^2+\sigma_2^2)\,\omega^2/2}.$$

This is the Fourier transform of $\mathcal{N}(0, \sigma_1^2 + \sigma_2^2)$. Since the Fourier transform is unique,

$$\boxed{f_Z = \mathcal{N}(0,\, \sigma_1^2 + \sigma_2^2).} \qquad \blacksquare$$

Non-zero means

For $X \sim \mathcal{N}(\mu, \sigma^2)$, shifting by μ multiplies the Fourier transform by $e^{-i\omega\mu}$:

$$\mathcal{F}\{\mathcal{N}(\mu, \sigma^2)\}(\omega) = e^{-i\omega\mu}\, e^{-\sigma^2\omega^2/2}.$$

For $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ independent,

$$\mathcal{F}\{f_Z\}(\omega) = e^{-i\omega(\mu_1+\mu_2)}\, e^{-(\sigma_1^2+\sigma_2^2)\,\omega^2/2},$$

so

$$\boxed{X + Y \sim \mathcal{N}(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2).}$$

The means add and the variances add.

Covariance of Independent Random Variables

Let X and Y be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. We show that if X and Y are statistically independent, then their covariance is zero.

Step 1: $E\big[X - E[X]\big] = 0$

Since $E[X] = \mu_X$ is a constant, by the definition of expectation,

$$\begin{aligned} E\big[X - E[X]\big] &= \int_{-\infty}^{\infty} (x - \mu_X)\, f_X(x)\, dx \\ &= \int_{-\infty}^{\infty} x\, f_X(x)\, dx \;-\; \mu_X \int_{-\infty}^{\infty} f_X(x)\, dx \\ &= \mu_X - \mu_X \cdot 1 \\ &= 0, \end{aligned}$$

where we used $\int_{-\infty}^{\infty} x\, f_X(x)\, dx = E[X] = \mu_X$ and $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$ (a PDF integrates to 1). By the same argument, $E\big[Y - E[Y]\big] = 0$.

Step 2: $C_{XY} = 0$ **for independent** X **and** Y

By the definition of covariance and of the expectation of a function of two random variables,

$$C_{XY} = E\big[(X - \mu_X)(Y - \mu_Y)\big] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y)\, f_{XY}(x, y)\, dx\, dy.$$

If X and Y are statistically independent, their joint PDF factors into the product of the marginal PDFs:

$$f_{XY}(x, y) = f_X(x)\, f_Y(y).$$

Substituting, the integrand becomes a function of x times a function of y, so the double integral separates into a product of two single integrals:

$$\begin{aligned} C_{XY} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)\, f_X(x)\, (y - \mu_Y)\, f_Y(y)\, dx\, dy \\ &= \left[\int_{-\infty}^{\infty} (x - \mu_X)\, f_X(x)\, dx\right] \left[\int_{-\infty}^{\infty} (y - \mu_Y)\, f_Y(y)\, dy\right] \\ &= E\big[X - E[X]\big]\; E\big[Y - E[Y]\big]. \end{aligned}$$

By Step 1, both factors are zero, so

$$\boxed{C_{XY} = 0 \cdot 0 = 0.} \qquad \blacksquare$$